MINOR PROJECT 2026–27 · DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

JASIPA

Job Applicant Screening and Interview Panel Agent

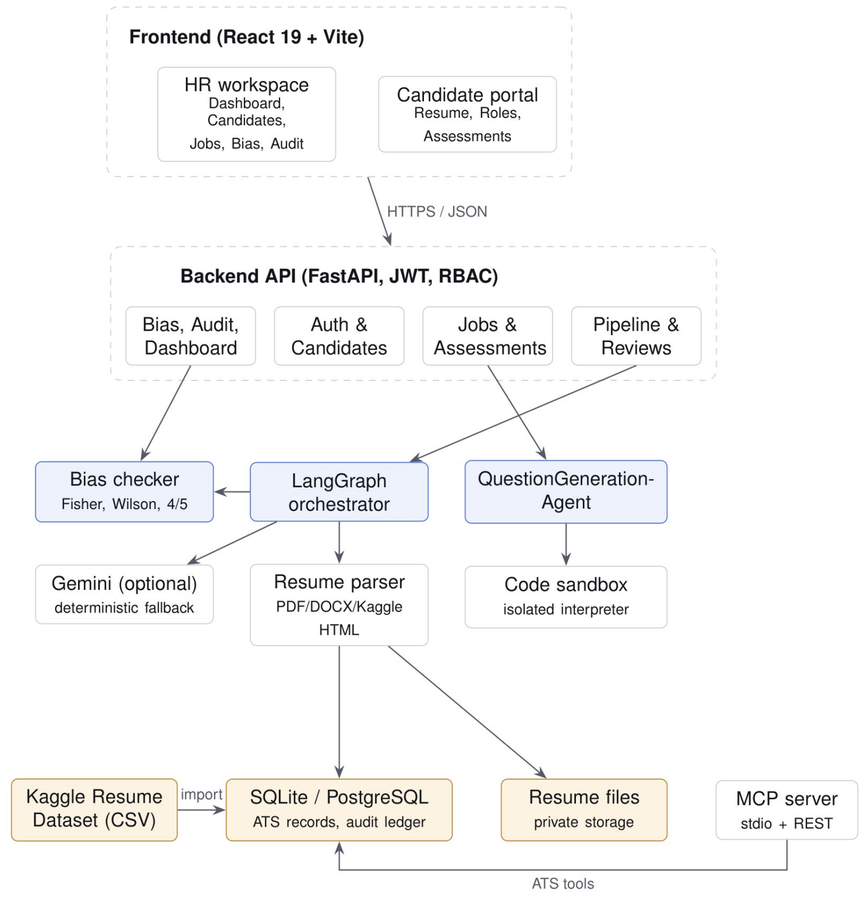
[Team member names] · Guide: [Guide name]

Three independent agents screen, assess and evaluate each candidate; a coordinator merges their evidence; every recommendation carries a bias check and goes to a human, who records the decision in a verifiable audit trail.

Problem

- Manual screening is slow and inconsistent between reviewers and rounds.
- Small applicant pools make adverse-impact checks easy to misread — no data is often reported as no bias.
- Recruitment AI is high-risk under the EU AI Act; automated rejection without human review is unacceptable.

Architecture



Bias check: three honest states

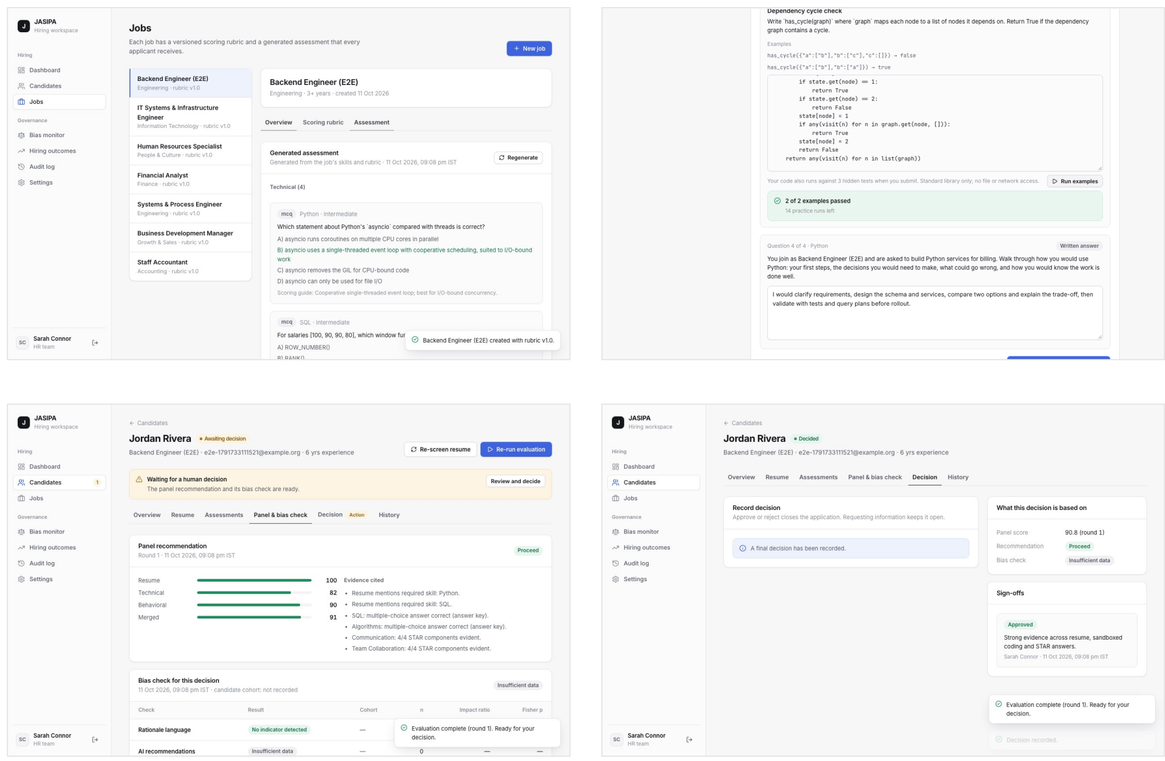
Disparity detected
Impact ratio < 0.80 with n ≥ 5 per cohort

No indicator detected
Screen not triggered; significance is a note

Insufficient data
Too few candidates — never shown as fair

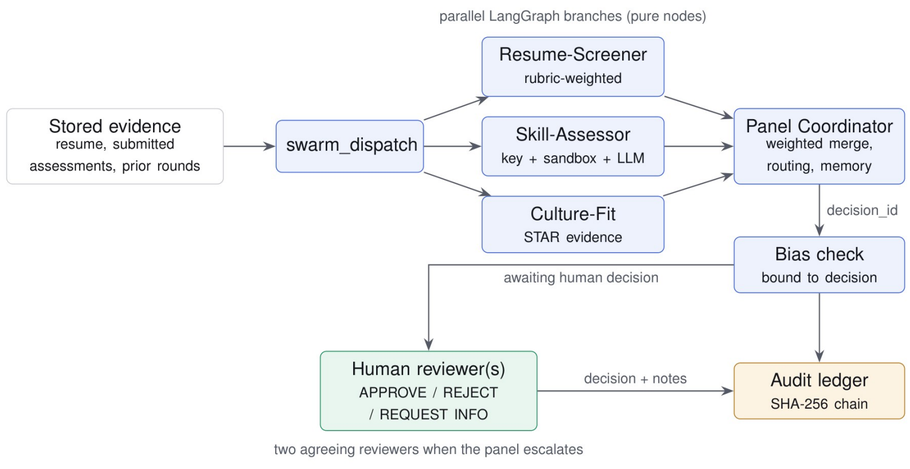
Fisher's exact test and Wilson 95% intervals quantify uncertainty. Cohorts are HR-recorded labels, never inferred and never shown to agents. The check informs the reviewer; it never changes scores or routing.

System in use



Generated assessment · sandboxed coding · panel with bias check · human decision

Agentic workflow



- **Tools:** resume_screen · skill_assess (sandboxed code) · bias_check · MCP ATS tools
- **Skill:** versioned, JD-anchored rubric with stage weights 0.4 / 0.4 / 0.2
- **Escalation:** two reviewers when agents diverge by ≥ 30 points

Data and validation

2,484 Kaggle resumes

48 imported (6 roles)

149/149 tests passed

Suite	Total	Pass	Fail
Backend (pytest)	110	110	0
Frontend (Vitest)	30	30	0
End-to-end (Playwright)	9	9	0

- Real Kaggle resumes replace synthetic seed data; dataset records are labelled and kept apart from applicants.
- Test tables are generated from the runners' result files — unit, API, component and 9 end-to-end journeys.

Does it learn?

No. Scoring is rubric-based and inspectable; an optional LLM helps with open answers, questions and summaries. Outcomes are stored for analytics only — learning from past hiring risks encoding past bias.

Limitations · next steps

No real cohort data or hiring outcomes; LLM output quality not evaluated; subprocess sandbox is defence in depth. Next: audited rubric updates, expert-agreement validation, container sandbox by default.

github.com/UditaB20/jasipa